Rajagopal Somaskandan

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Professional Summary

M.Sc. Computer Science candidate (Sathyabama, 2027) and published IEEE author who ships full-stack AI systems — not prototypes. Proven across fraud detection (94% recall, ROC-AUC 0.995), multilingual NLP, real-time computer vision, and 3D scene reconstruction using Gaussian Splatting. Two conference publications, a national hackathon win, and a production internship at an AI startup. Builds end-to-end: model training on GPU, FastAPI deployment, and browser-ready interfaces — without waiting for a team.

Technical Skills

Languages: Python, Java, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, SHAP, Feature Engineering, Model Evaluation
NLP & GenAI: BERT, Sentence-Transformers, spaCy, NLTK, LangChain, ChromaDB, FAISS, RAG, Prompt Engineering
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB, Redis, SQLite
Frameworks & Tools: FastAPI, Flask, Streamlit, Docker, Git, GitHub, MLflow, Netlify, Render
Data Analysis: Pandas, NumPy, Matplotlib, Plotly, Power BI

Education

Sathyabama Institute of Science and Technology — **B.Sc. Computer Science (AI Specialization)** | First Class — *2025*
Sathyabama Institute of Science and Technology — **M.Sc. Computer Science** | — *Expected 2027*

Experience

Production AI Developer (Intern) | *Innoxr Labs Pvt. Ltd.* — *Mar 2026 – Jun 2026*

- Designed and shipped applied ML pipelines and explainable AI features for flagship deep-tech products, directly contributing to the core product roadmap in a fast-moving startup environment.
- Architected and deployed production FastAPI microservices integrating trained ML models, reducing inference latency and enabling scalable API-first access to AI capabilities.
- Drove end-to-end feature ownership — from model iteration to deployment — across immersive AI and digital solution verticals, shipping production-grade code under live constraints.

Data Analyst Intern | *India Advocacy* — *Jul 2024 – Oct 2024*

- Engineered automated web-scraping pipelines (Java + Selenium) to extract, clean, and structure ***1M+ raw records***, distilling them into 15–20k curated, campaign-ready datasets.
- Implemented multi-stage validation and data-quality controls, reducing downstream data errors and ensuring business-usability standards for targeted outreach campaigns.

Security Trooper (National Service) | *Singapore Armed Forces* — *Jan 2020 – Jan 2022*

- Executed structured security operations and supported the Ministry of Health COVID-19 task force through data collection, field reporting, and operational coordination.

Projects

MonoSplat: Single-Video 3D Gaussian Splat Reconstruction | *Python, PyTorch, COLMAP, FFmpeg, FastAPI, Three.js* — *2026*

- Built an end-to-end pipeline that converts a single phone video into a photorealistic, browser-navigable 3D Gaussian Splat scene, eliminating the need for depth sensors, multi-camera rigs, or desktop rendering software.
- Designed a CPU/GPU split architecture: FFmpeg frame extraction and COLMAP SfM pose estimation run locally; a custom PyTorch `GaussianTrainer` trains on Colab T4 GPU (20–40 min) using L1 + SSIM photometric loss with adaptive Gaussian densification and pruning.
- Shipped a full-stack web portal (FastAPI + Three.js) with SSE live job tracking, automated sequential COLMAP matching, and browser-optimized `.splat` binary export achieving ***30+ FPS*** real-time rendering.

FinGuard Pro: Explainable AI for Financial Fraud Detection | *Python, XGBoost, SHAP, FastAPI, SQL* *2025*

- Trained an XGBoost fraud detection model on imbalanced UPI-style transaction data, achieving ***94% recall*** and ***ROC-AUC 0.995*** — published at ICAAIC 2025 (ISBN: 979-8-3315-6587-9).
- Cut false positives by layering XGBoost predictions with rule-based AML heuristics targeting velocity abuse and structuring patterns, improving precision without sacrificing recall.
- Integrated SHAP for per-transaction explainability, producing audit-ready outputs aligned with financial regulatory compliance requirements.

S.A.F.E: Real-Time Crowd Anomaly & Risk Detection | *OpenCV, Farneback Optical Flow, MAD, FastAPI, Streamlit* *2026*

- Architected a privacy-preserving anomaly detection system using zone-based optical flow aggregation, achieving ***ROC-AUC 0.8391*** on a cross-domain Mall dataset with ***zero retraining***.
- Reduced false positives by ***40–60%*** via global motion compensation for camera shake on handheld and drone footage, eliminating a critical failure mode for field deployment.
- Benchmarked 5 unsupervised detectors (MAD, Z-Score, Isolation Forest, One-Class SVM, LSTM Autoencoder) and selected MAD for superior cross-domain generalization on out-of-distribution data.

Redline: Multilingual Preventive Healthcare Chatbot | *IndicBERT, FastAPI, Groq LLaMA 3.3, Telegram Bot API* *2026*

- Fine-tuned IndicBERT across 10+ Indian languages for intent classification, achieving ***82.4% accuracy*** on real-world health queries at the edge of low-resource NLP.
- Engineered a confidence-aware hybrid routing engine with Groq LLM fallback, attaining ***95.6% safety filter accuracy*** against harmful or out-of-scope medical responses.
- Deployed a three-tier medical escalation system with real-time outbreak alerts across all 28 Indian states via Telegram, serving a live multilingual user base.

CareerGenie: AI-Powered Job Matching & Skill Gap Analysis | *Claude API, Sentence-Transformers, ChromaDB, FastAPI, React2026*

- Built a full-stack RAG pipeline pairing Sentence-Transformers (all-MiniLM-L6-v2) with ChromaDB to semantically match resumes against live job listings scraped via SerpAPI, delivering ***granular 0–100% match scoring*** per role.
- Designed a hybrid scoring engine fusing vector similarity with skill-overlap analysis; routed results to Claude API to generate personalized match explanations and actionable skill gap reports.
- Won ***1st Place*** and ***Best Industry Innovation Award*** at Industry Innovation ’26 (national level, 63+ teams) with this platform.

SourceUp: Constraint-Aware Explainable Supplier Recommendation System | *Sentence-BERT, FAISS, FastAPI, Scikit-learn, Java (Jsoup*

- Designed a constraint-aware supplier ranking framework integrating budget, delivery time, and reliability directly into scoring, replacing traditional filter-then-rank pipelines.
- Built a semantic retrieval system using Sentence-BERT + FAISS over scraped B2B supplier data, enabling natural language-based supplier discovery.
- Developed an explainability layer providing feature-level contributions and trade-off insights (cost vs delivery vs rating) for transparent decision-making.
- Engineered an end-to-end pipeline (scraping → feature engineering → ranking → API) using FastAPI and Java-based data collection modules.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* *ISBN: 979-8-3315-6587-9*

Real-Time Explainable Fake News Detection using BERT-LSTM | *ICASET 2025*

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology — National Level* *Mar 2026*

- Won ***1st Place*** and ***Best Industry Innovation Award*** at Industry Innovation ’26 (organized with ATRIBS and Bluewind Innovations) — defeating 63+ teams from institutions across India with the CareerGenie AI platform.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* *Apr 2026*

- Competed in a 24-hour sprint organized by the SIMATS Hackathon Club, Saveetha Institute of Medical and Technical Sciences.

Certifications

AI and Machine Learning | *SpaceZee Technologies* *Jun 2024*

Python Core | *SpaceZee Technologies* *Feb 2024*